AI NOTES SERIES

AI Agentic Workflows

How AI Agents Plan, Use Tools, and Finish Multi-Step Work

Simple, clear notes for teachers, students, and curious beginners. What an AI agent is, how its working loop runs, why tools, memory, and permissions matter, and where agents still fail - explained in plain words, with no technical background needed.

These are study notes written from hands-on familiarity - from daily work with an AI assistant - not a technical manual.

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THE BASICS

What an AI Agent Is

An AI agent is an AI system that is given a goal and then works toward that goal in steps, instead of only answering one question.

A normal chatbot answers and stops. You ask a question, it gives a reply, and the work is done. An agent works differently. Given a goal, it breaks the task into smaller steps, chooses tools to do each step, carries the steps out, looks at the results, corrects its course when something is wrong, and reports back at the end.

Chatbot and agent, side by side

- **A chatbot replies.** It works one message at a time, mostly with words, and it waits for you after every answer.
- **An agent acts.** It works through a plan, uses tools between replies, and keeps going until the goal is reached or it is stopped.
- **A chatbot forgets the plan.** Each new question starts almost fresh, unless you repeat the background.
- **An agent carries the plan.** It remembers what it has already done and what still remains to be done.

A simple example

Imagine the goal: **prepare a short report about solar panels for a school**. A chatbot would write one answer from what it already knows. An agent would first plan the work, then search for current information, read the most useful sources, note the key facts, draft the report, check the draft against the sources, fix any mistakes, and finally hand you the report with a short summary of what it did.

> **A simple way to remember it:** a chatbot is like asking a knowledgeable friend one question. An agent is like giving a careful assistant a job, the tools for the job, and asking for a report when the job is finished.

The word **agentic** simply means working in this agent-like way: goal first, then planned steps, tool use, checking, and a final report.

HOW IT WORKS

The Loop, in Words

Every agent run follows the same basic loop. The stages below are the whole workflow in one line - read it first, then read what each stage means.



Goal

The run begins with a clear goal from the user: what should be achieved, for whom, and with what limits. Every later decision is judged against this goal, so a vague goal usually leads to a vague result.

Planning

The agent breaks the goal into an ordered list of smaller steps. Good planning names what must happen first, what depends on earlier results, and what a finished job looks like. The plan can change if new information appears.

Tool selection

For each step, the agent chooses a suitable tool - search for finding information, a file tool for reading a document, or a code tool for numbers. Choosing the right tool for each step is a large part of doing the job well.

Execution

The agent now carries out the steps: it runs the searches, reads the files, does the calculations, and produces drafts or results. This is where the real work happens - and where real mistakes can happen.

Evaluation

After each important step, the agent checks the result: Is this correct? Is anything missing? Does it still match the goal? If the check fails, it repeats the step, tries another tool, or changes the plan. This checking separates an agent from a simple automatic script.

Response

When the goal is reached, the agent reports back: what it did, what it found, and what the final result is. A good report is honest about problems and gaps, not only about successes.

> **Remember:** evaluation often sends the agent back to planning or execution, so the loop may turn several times before the final response.

THE WORKING PARTS

Tools and Memory

An agent without tools can only talk. Tools let it act, and memory lets it keep track of its own work.

The kinds of tools an agent can use

- **Search and reading tools** - find information on the web or in documents, and read the most useful sources in full.
- **File tools** - open, read, create, and edit documents, spreadsheets, images, and other files.
- **Code and calculation tools** - run small programs, do accurate arithmetic, sort and clean data, and convert files between formats.
- **Other services** - connected apps and services, such as email, calendars, or design tools, which the agent can use within the limits it is given.
- **Communication tools** - prepare drafts, summaries, and messages for the user to review and send.

The exact tools differ from one agent system to another. What matters is the idea: the language model does the thinking and deciding, and the tools do the reaching, reading, calculating, and making.

How memory helps

A multi-step job creates a problem: by the fifth step, the agent must still know what the first four steps found. Memory solves this. It carries the goal, the plan, earlier results, and important facts from one step to the next, so the work stays connected instead of starting again at every step.

- **Working memory** holds what is needed during the current run - the goal, the plan, and the results so far.
- **Saved notes and files** let longer jobs keep their findings in documents the agent can re-read later, even in a new session.
- **User context** - facts and preferences the user has shared - helps the agent make suitable choices without asking the same questions again.

> **Honest point:** memory is useful, but it is not perfect. An agent can carry a wrong fact forward just as easily as a right one, which is why checking at each step matters - the next chapter and the limits chapter explain this.

SAFETY FIRST

Why Permissions Matter

An agent that can act can also act wrongly - and it can act wrongly very fast, many times, before anyone notices.

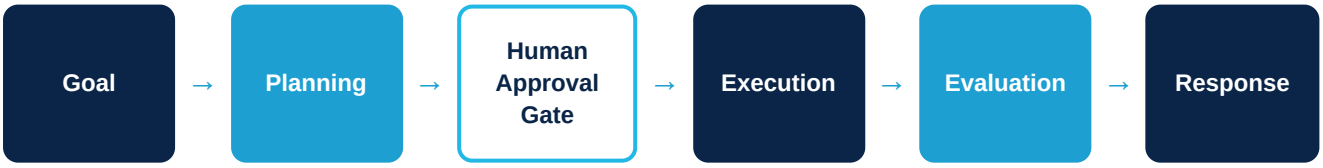
This is the most important safety idea in these notes. A chatbot that gives a wrong answer has only produced wrong words, which you can ignore. An agent that takes a wrong action may have sent a message, changed a file, or deleted something. Safe agent design therefore controls what the agent is allowed to do.

The four safety habits

- **Limited tool permissions.** Give the agent only the tools and access the current job needs - nothing more. A research job does not need permission to send messages or delete files.
- **Clear boundaries.** State plainly what the agent must never do: which files are off limits, which actions are forbidden, and which topics or people are out of scope.
- **Logging.** Keep a record of what the agent did at each step - which tools it used and what changed - so a human can review the run afterwards and find where a problem began.
- **Human approval before sensitive steps.** Before any step that is irreversible or sensitive - sending, publishing, paying, deleting, or sharing private information - the agent must stop and wait for a human to approve.

Where the approval gate belongs

The approval gate sits inside the loop, between planning a sensitive action and executing it. The agent may plan freely and do safe, reversible work on its own, but it pauses at the gate until a person says yes.



> **Golden rule:** the more an action cannot be undone, the stronger the human check must be before it. Reversible work can run freely; irreversible work waits for a person.

IN PRACTICE

What Agents Are Used For

Agents earn their place on work that is too long, too detailed, or too step-heavy for a single prompt.

A single prompt works well for a single answer: one explanation, one paragraph, one short list. But many real jobs are chains - find this, then read that, then compare, then draft, then check, then deliver. That chain is exactly what an agent is built to carry.

Common uses

- **Multi-step research.** Searching several sources, reading the best ones in full, comparing what they say, and preparing a grounded summary with the sources named.
- **Document processing.** Reading long documents, extracting the important points, reorganising them, and producing clean new documents - summaries, reports, tables, or study notes like these.
- **Data tasks.** Cleaning messy lists, sorting and grouping records, doing calculations with code tools, and presenting the results in a clear table.
- **Productivity workflows.** Preparing drafts, organising files, updating a website, checking a long piece of work page by page - jobs made of many small steps that must all be done in the right order.
- **Learning support.** Building practice material step by step: first the topic outline, then examples, then questions, then an answer key - each stage checked before the next begins.

When a plain chatbot is enough

Not every job needs an agent. For a quick explanation, a translation, or help wording one paragraph, a chatbot is faster and simpler. Use an agent when the work has many steps, needs tools, or must be checked as it goes - and keep a human in charge of the goal and the final approval.

> **From daily practice:** the pattern that works best is simple - the human sets the goal and the boundaries, the agent does the long chain of steps, and the human reviews the report and approves anything sensitive.

USE IT WISELY

Honest Limits

Agents are powerful, but a long chain of steps is also a long chain of chances to go wrong.

- **Errors compound across steps.** A small mistake in an early step becomes the base for every later step. A wrong figure found in step two can quietly poison the table in step five and the conclusion in step eight.
- **A wrong early assumption can steer a whole run.** If the agent misunderstands the goal, or trusts a poor source at the start, it may work hard and efficiently - in the wrong direction - for the entire job.
- **Evaluation is needed at each stage, not only at the end.** Checking only the final result is too late: by then, early errors are buried deep inside the work. Checking step by step catches mistakes while they are still cheap to fix.
- **Confidence is not proof.** An agent can report a wrong result in a calm, confident tone, just as a chatbot can. A tidy report is not evidence that the work is correct.
- **Tools can fail or mislead.** A search can return weak sources, a file can be outdated, and a service can be unavailable. A good agent reports these problems instead of quietly working around them with guesses.
- **A human must actually read the report.** The safety system fails if the person approves everything without reading. The report, the log, and the approval gate only protect you if a human genuinely checks them.

> **Golden rule:** trust an agent the way you would trust a fast, tireless junior assistant - give it real work, but check the work, especially anything important, surprising, or impossible to undo.

Good habits when you use an agent

- **State the goal and the limits clearly** before the run starts, including what the agent must not do.
- **Ask for sources and a step-by-step report**, so you can verify the important claims yourself.
- **Keep approval gates on** for sending, publishing, deleting, paying, and sharing private information.
- **Spot-check the output** against original sources, even when the report looks polished.

REFERENCE

Words to Know

Word	What it means, in plain words
Agent	An AI system given a goal that plans steps, uses tools, checks results, and reports back.
Goal	The result the user wants the agent to achieve, with its limits and boundaries.
Planning	Breaking the goal into an ordered list of smaller steps before the work begins.
Tool	A function the agent can use to act - such as search, reading files, running code, or using a connected service.
Execution	Carrying out the planned steps with the chosen tools.
Evaluation	Checking a step's result against the goal before moving on, and correcting course when needed.
Memory	The information an agent carries across steps - the goal, the plan, and results so far.
Permission	What an agent is allowed to do: which tools it may use and which actions it may take.
Approval Gate	A planned pause where the agent must wait for human approval before a sensitive or irreversible step.

Keep this page as a quick reference. Every term here appears in the earlier chapters, where it is explained with examples.

> **How to use these words:** when you read about a new AI tool, ask which of these parts it has - a goal, a plan, tools, memory, evaluation, and an approval gate. The answers tell you quickly how agent-like it really is, and how safely it is designed.

FINAL PAGE

Quick Revision

Read this box before a test, a presentation, or whenever you want the whole topic in one view.

The most important points

- An **agent** works toward a **goal** in steps; a chatbot only replies and stops.
- The loop is **Goal, Planning, Tool Selection, Execution, Evaluation, Response** - and evaluation can send the work back for another turn.
- **Tools** let the agent search, read, calculate, and create; **memory** carries context across the steps.
- Safe design means **limited permissions**, clear boundaries, logging, and a **human approval gate** before sensitive or irreversible steps.
- Agents suit **long, multi-step work**: research, documents, data tasks, and productivity workflows.
- **Errors compound** across steps, so check at each stage - and a human must truly read the final report.

A closing note

These notes are introductory study notes, and the explanations are simplified on purpose. Keep exploring the topic in depth, and use every new tool with wisdom and honesty.

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